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Precision Enhancement in North-South Directional Drilling Through Definitive Dynamic Surveys

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The conventional approach of using continuous direction and inclination while drilling the wellbore was developed and implemented in the late 1970s and became widely adopted in the drilling industry in the early 2000s. This revolutionary method helped the operators to step up in well delivery from drilling only vertical wells to reaching reservoir with application of deviated wells. It became a landmark in directional drilling history, as continuous data was used to steer the well proactively. This allows directional drilling specialists to determine the exact position of the wellbore at any given moment and take faster decisions for trajectory corrections during drilling. The technology has demonstrated good reliability and is now an industry standard.

However, this conventional method, which relies on the projection of rotational components and static surveys as references, has limitations when determining the precise direction of the wellbore. Drilling wellbores in the magnetic North-South direction pose a significant technical challenge for conventional Measurement While Drilling (MWD) tools, which use standard Direction and Inclination (D&I) surveying packages to acquire accurate continuous Azimuth data. These tools are relying on Earth's magnetic field to determine orientation. And in direction close to the magnetic poles MWD tool will have reduced capability to resolve direction and provide reliable continuous magnetic Azimuth measurement.

Additionally, the magnetic field is prone to interference, further reducing the precision of conventional D&I packages. Drilling equipment and any nearby metallic components exacerbate these challenges, complicating direction determination and computation in such a complex magnetic field environment. Another challenge is cross axial interference, which can affect static survey and lead to inaccurate reference for MWD tool to compute continuous Azimuth.

Introduction

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While drilling within 20 deg proximity to the Magnetic North or South, conventional D&I packages are unable to obtain precise Azimuth readings, which can result in uncertainty in wellbore placement unless a static survey station is taken. This limitation can lead to a range of negative consequences, including degradation of the wellbore quality, severe cumulative tortuosity, increased open hole friction factor, difficulties in installing completions tubing, and challenges in performing wireline logging.

In today's market, more operators are focusing on developing mature projects and brown fields. Considering the complexity of the fields, these projects represent congested platforms and numerous wells drilled in various directions to delineate the boundaries of reservoirs, adding further complexity to drilling operations. As a result, many projects may face scenarios where drilling wells in the North-South direction becomes unavoidable. Given the increasing complexity and the demand for advanced technologies and methodologies, there is a growing need for a new approach to define and manage the wellbore path.

Fundamentals of the conventional measurement

During continuous surveys, axial magnetometers and accelerometers (B_x and G_x) are recorded continuously, while cross-axial components (B_y , B_z , G_y , G_z) are reused from the last accepted static survey. This hybrid method uses six axes — B_x and G_x from the current readings, and B_y , B_z , G_y , G_z from the previous static survey — to calculate inclination and Azimuth during drilling.

While this approach usually provides acceptable results and allows continuous update of tool position, it has a limitation. Cross-axial sensor values can change during drilling and reusing them from previous station means introducing an error due to dynamic variations. This error becomes most significant when drilling in the North – South direction. According to the ISCWSA Error Model (1), the weight function for cross-axial magnetometer error terms becomes important when Azimuth is close to 0 or 180 degrees.

The degradation of Continuous Azimuth accuracy in North-South direction has been illustrated on a real-field data from the well in Qatar. For each Static Survey (SS) station $i = 1, 2, \dots$, positioned approximately 30 meters apart, the Continuous Survey (CS) Azimuth was calculated by using newly update accelerometer $G_x^{SS,i}$ and magnetometer $B_x^{SS,i}$ axial readings and obsolete cross-axial readings from the previous SS station $i-1$, see Figure 1. Hence, the estimated CS Azimuth is a function of

$$Az_{est}^{CS,i} = f(G_x^{SS,i}, G_y^{SS,i-1}, G_z^{SS,i-1}, B_x^{SS,i-1}, B_y^{SS,i-1}, B_z^{SS,i-1})$$

Static Survey is acquired in a stable environment (no dynamic noise or errors due to reusing obsolete sensor values), hence at the chosen station i , SS Azimuth is considered the ground truth. The introduced CS Azimuth error at the same position due to reusing obsolete cross-axial sensor readings, has been estimated for each station i by

$$e_i^{CS} = |Az_i^{SS} - Az_i^{CS}|$$

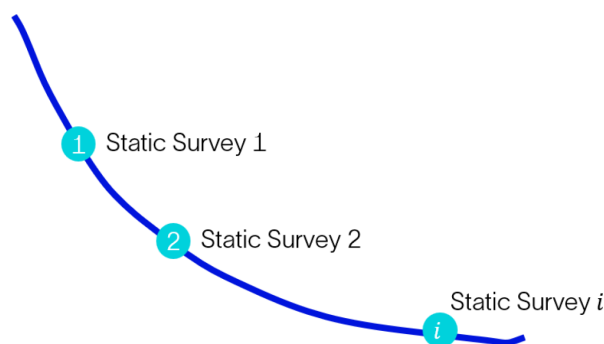


Figure 1—Example of data selection

As shown in Table 1 below, the error e_i^{CS} is significantly more important in North-South direction. That means that Continuous Survey has degraded quality at the magnetic North/South.

Table 1—Error of continuous surveys is bigger in North-South direction

Depth	Mag Azm	Gx	Gy	Gz		Bx	By	Bz	
3002.91	1.06	478.62	-868.61	64.07		44807.81	-2319.10	-537.18	Static Survey
3002.91	357.59	478.62	-330.67	-798.30		44807.81	-2104.80	-852.93	Continuous Survey at Static MD
	3.47								$Az_1^{CS} - Az_1^{SS}$
6955.92	278.69	283.00	-806.64	-510.80		6964.66	-22083.02	18780.53	Static Survey
6985.30	278.76	286.45	-806.64	-510.80		7279.72	-22083.02	18780.53	Continuous Survey at Static md
	0.07								$Az_1^{CS} - Az_1^{SS}$

The CS Azimuth accuracy will deteriorate gradually when approaching magnetic Azimuth = 0/360 or Azimuth = 180 degrees. It was observed that within 5 degrees from these poles Continuous Survey is not reliable, as illustrated by the red zone in Figure 2. In this red area Azimuth error may reach up to 5 degrees, which creates high uncertainty for the directional drilling.

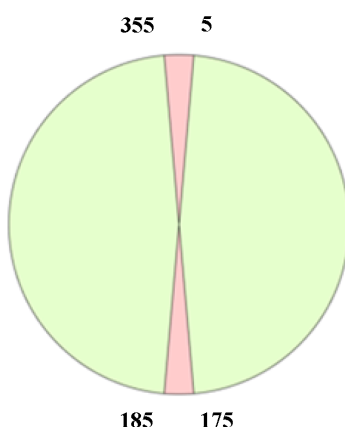


Figure 2—Zone of reduced accuracy for Continuous Surveys

The most critical scenario arises in horizontal wells (with Inclination close to 90 deg), drilled along a North–South trajectory. Due to operational constraints, adjusting the drilling direction in such wells is difficult. As a result, extended horizontal sections in the North–South direction are particularly susceptible to cumulative positioning errors.

This issue can be effectively mitigated using standard industry practices. One common approach involves acquiring Static Surveys after each connection (approximately every 100 feet or 30 meters), which provides updated six-axis sensor data. These measurements can then be used to recalibrate the Continuous Survey and maintain Azimuth accuracy. In critical intervals, additional intermediate Static Surveys may be conducted to further reduce Azimuth uncertainty and enhance trajectory control.

However, this approach has several drawbacks. Firstly, it requires to stop the drilling process to acquire the SS in the stable environment. This procedure increases both drilling time and cost. Secondly, pauses in tool rotation and circulation may introduce additional wellbore stability risks. Finally, SS can be affected by cross-axial magnetic interference, i.e. bias and scale factor errors in cross-axial magnetometer measurements. This can happen for example, when MWD tool in the BHA is close to metallic components. In this case the reliability of the entire well trajectory would be degraded, as it is refined based on compromised SS data. On one hand, hybrid Continuous Surveys carry inherent limitations in measurement accuracy. On the other hand, frequent reliance on Static Surveys increases drilling time, operational costs, and associated risks. A compelling solution lies in the adoption of technology that delivers real-time, full 6-axis measurements and eliminating the need for reused data or interruptions in drilling. Definitive Dynamic Surveys (DDS) allow continuous update of all 6 axes, resulting in accurate and high-frequency MWD surveys. DDS represent a promising advancement, offering continuous, high-quality trajectory data without compromising efficiency or safety.

Definitive Dynamic Survey theory

Definitive Dynamic Surveys (DDS), also referred to as Dynamic Magnetic Surveys (DMS), began development in the mid-2010s and saw widespread commercial adoption in the early 2020s. This technology enables the acquisition of high-quality, continuously updated MWD surveys while drilling, eliminating the need to stop operations for static surveys. This advancement significantly reduces rig time associated with surveying, lowers drilling risks by avoiding additional pump-off periods, and improves true vertical depth (TVD) accuracy and directional control.

DDS surveys are collected at a high frequency during drilling operations, eliminating the delays and procedures traditionally required to obtain good-quality Static MWD surveys. Like Static Surveys, DDS relies on full 6-axes sensor updates: three orthogonally arranged magnetometers and three accelerometers. However, DDS is deployed on an enhanced survey platform featuring advanced hardware and firmware improvements over traditional MWD systems: modern sensor technologies and enhanced computational resources allow multiple opportunities for better signal processing. Moreover, DDS algorithms allow to compensate for multiple error sources, including dynamic errors, which occur while drilling:

- Cross-axial magnetic interference,
- Eddy current errors
- Shock and vibration.

These improvements allow DDS to produce surveys of comparable or superior quality to conventional Static MWD surveys. The underlying design and error modeling of DDS technology have been thoroughly described in technical literature, including works by [Lowdon et al. \(2019\)](#) (2), [ElGizawy et al. \(2019\)](#) (3), [ElGizawy et al. \(2020\)](#) (4) and other related works (5).

As previously discussed, the traditional Continuous Survey approach suffers from degraded Azimuth accuracy when drilling in magnetic North or South directions. This issue stems from the partial reuse of sensor data and becomes critical in long, near-horizontal sections where precise Azimuth control is essential.

DDS technology effectively overcomes this limitation by delivering full real-time 6-axis data at high frequency, eliminating the need for interpolation or reuse of prior survey measurements. Matching the accuracy of Static Surveys, DDS maintains high Azimuth fidelity even in magnetically challenging directions. This makes it a robust and dependable solution for precise trajectory control, particularly in North–South drilling applications.

The following section presents real-world case studies demonstrating how DDS eliminates Azimuthal uncertainty in magnetic North–South drilling environments—delivering a step-change in both survey accuracy and operational efficiency.

Implications for Directional Well Delivery

Magnetic MWD tools determine Azimuth by using Earth's magnetic and gravity fields. When drilling the wellbore in a trajectory oriented close to magnetic North or magnetic South, the magnitude of the cross-axis (perpendicular) component of the magnetic field, as measured by the sensor, becomes extremely small or nearly zero. In this situation, even minor errors, sensor noise, or disturbances in the cross-axis measurement will have a significant amplified effect on the calculated Azimuth value. This results in Azimuth readings becoming highly sensitive and unstable.

Practically, the sensor attempts to resolve the direction of horizontal magnetic field, but when tool axis is almost perfectly aligned with North or South, the signal in cross-axis is minimal, almost not present. In this condition, even very small noise or magnetic interference can cause large error in Azimuth output, similar like using compass needle that already points North — small mechanical shock or external field can easily move the reading far from true direction.

The following challenges have been consistently observed during North-South drilling operations:

1. Unreliable Continuous Azimuth in North-South Direction

- When drilling in a North-South orientation, the horizontal component of the Earth's magnetic field vector becomes minimal. This directly impacts magnetic MWD tools:
- The magnetic field vector aligns closely with the borehole axis, making it hard for the tool to resolve Azimuthal changes accurately.
- Small noise or sensor errors result in disproportionately large fluctuations in Azimuth readings.
- RSS or motor steering control becomes unreliable because toolface orientation cannot be trusted in real time.

2. Delay in Azimuthal Correction

Unreliable Azimuthal data leads to delays in identifying and correcting deviations from the planned trajectory. This lag increases the risk of wellbore tortuosity, missed targets, and elevated anti-collision risks in highly dense multi-well environments.

3. High Wellbore Tortuosity

Frequent reactive corrections—caused by delayed or poor Azimuth information—lead to a tortuous wellbore:

- Micro-doglegs accumulate across the lateral.
- Steering adjustments are made without confidence, creating unnecessary undulations.
- This directly results in:

- Higher torque and drag
 - Difficulty running liners or completions
 - Poor cement coverage, especially in openhole sections, compromising zonal isolation and long-term well integrity
4. Risk of Unsuccessful Liner Deployment
- Increased tortuosity and excessive dogleg severity create mechanical restrictions for running liners, especially toward the end of the lateral:
- Bending and friction forces increase dramatically.
 - Liner systems may hang up before reaching target depth, requiring costly contingency measures like:
 - Liner rotation tools
 - Reaming runs
 - Liner re-design
 - In severe cases, this results in liner failure or the need to sidetrack.
5. Risk of Breaching Lateral Tolerance Limits in Narrow Lateral Separation
- In multi-well pads with tight lateral spacing (e.g., producer-injector patterns), maintaining separation is crucial:
- In North-South wells, poor Azimuth control leads to high lateral uncertainty, especially in the horizontal section.
 - The risk of breaching minimum separation increases, potentially:
 - Triggering anti-collision alarms
 - Compromising reservoir management
 - Violating regulatory or operator-defined separation rules
6. Increase in Drilling Time Due to Stationary Surveys
- To compensate for unreliable Continuous Azimuth:
- Directional drillers pause more frequently to take stationary surveys and confirm actual wellbore direction.
 - These stops increase non-productive time (NPT) and slow down rate of penetration (ROP).
 - Over an entire lateral, this can translate into hours or days of rig time, affecting project schedule and cost.
7. Frequent Downlinking for Directional Corrections
- With continuous uncertainty in trajectory, drillers must:
- Downlink steering commands frequently to maintain course.
 - Each downlink introduces a pause, which cumulatively adds significant time to the operation.
 - Excessive downlinking also:
 - Raises the risk of Real-time communication issues, especially at high depths or in noisy environments

- Adds complexity to operations, particularly with RSS configurations where tool might have dozens of commands.

Resolution of Uncertainties and Strategic Advancements

The case study presented below is based on real application while drilling 3 offset platforms with reservoirs in the same direction (North-South), see [Figure 3](#). Platform A was drilled with a conventional practice in survey methodology. During execution phase a conservative approach was taken by controlling parameters that resulted in a below average Rate Of Penetration (ROP) that was achieved in this project. A significant increase in well tortuosity was observed due to inability to have accurate continuous Azimuth and relying on static survey stations. Platform B and C were drilled with new generation MWD technology and using DDS methodology for survey management and continuous Azimuth computation.

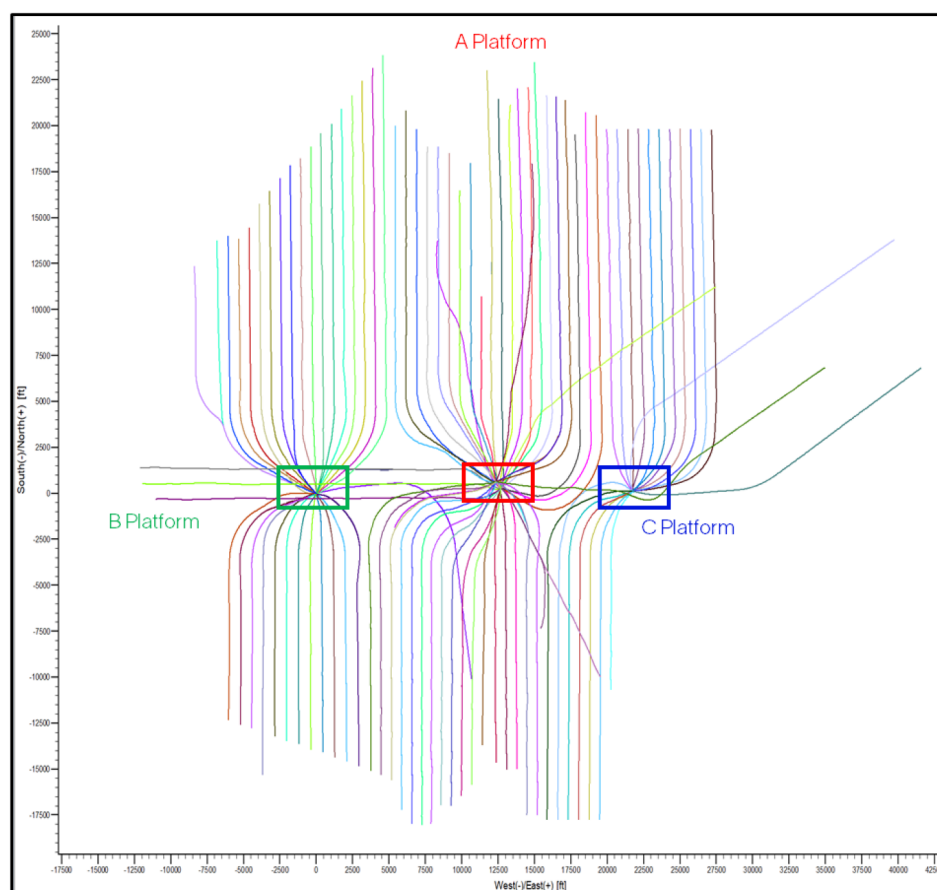


Figure 3—Plan view of 3 platforms. Platform A with a conventional MWD tool. Platform B and C were drilled with a new generation MWD tool by using DDS methodology

The integration of the new generation MWD tool with DDS surveying method in addition of high-speed telemetry system into directional drilling operations has significantly enhanced performance in North-South trajectories. This complex answer product enabled more reliable dynamic Azimuthal measurements, directly contributing to reduced wellbore tortuosity and lowering the risk associated with liner deployment in complex lateral sections. As shown in [Figure 4](#), DDS Azimuth (orange) readings are in good agreement with legacy Static Survey Azimuth (green), while Continuous Azimuth is scattered.

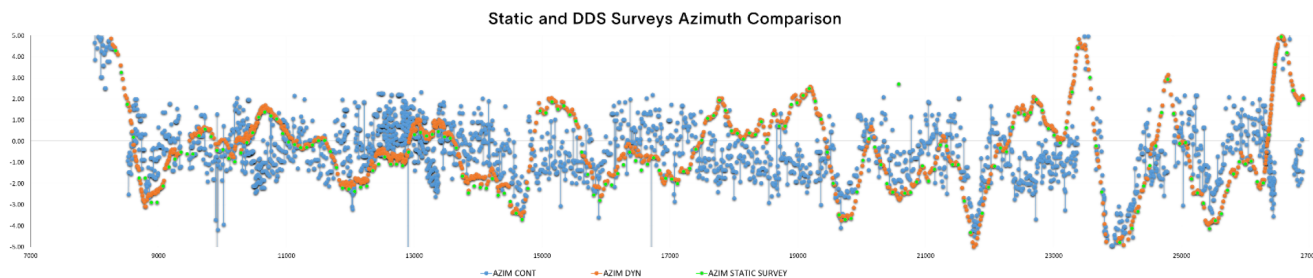


Figure 4—Example of the well as part of the case study. 3 survey measurements are presented on the graph: (a) Green color dots represent Azimuth from static surveys stations, (b) Orange color dots represent DDS Continuous Azimuth, (c) Blue color dots represent Conventional Continuous Azimuth.

The enhanced telemetry and real-time data transmission frame listing had to be adjusted to accommodate update of the continuous Azimuthal information without compromising Logging While Drilling data density. This enhancement has allowed for more precise trajectory control, minimizing deviation from planned paths and improving reservoir access. Continuous, accurate Azimuthal measurements in real time have also reduced the need for stationary surveys, leading to measurable reductions in non-productive time (NPT).

Drilling Performance Improvements

As illustrated in Figure 5, the initial deployment of the new generation MWD tool with DDS on new wells demonstrated immediate performance gains:

- 20% increase in On-Bottom Rate of Penetration (ROP)
- 22% increase in footage drilled per bit on bottom rotating time (FT/BRT)

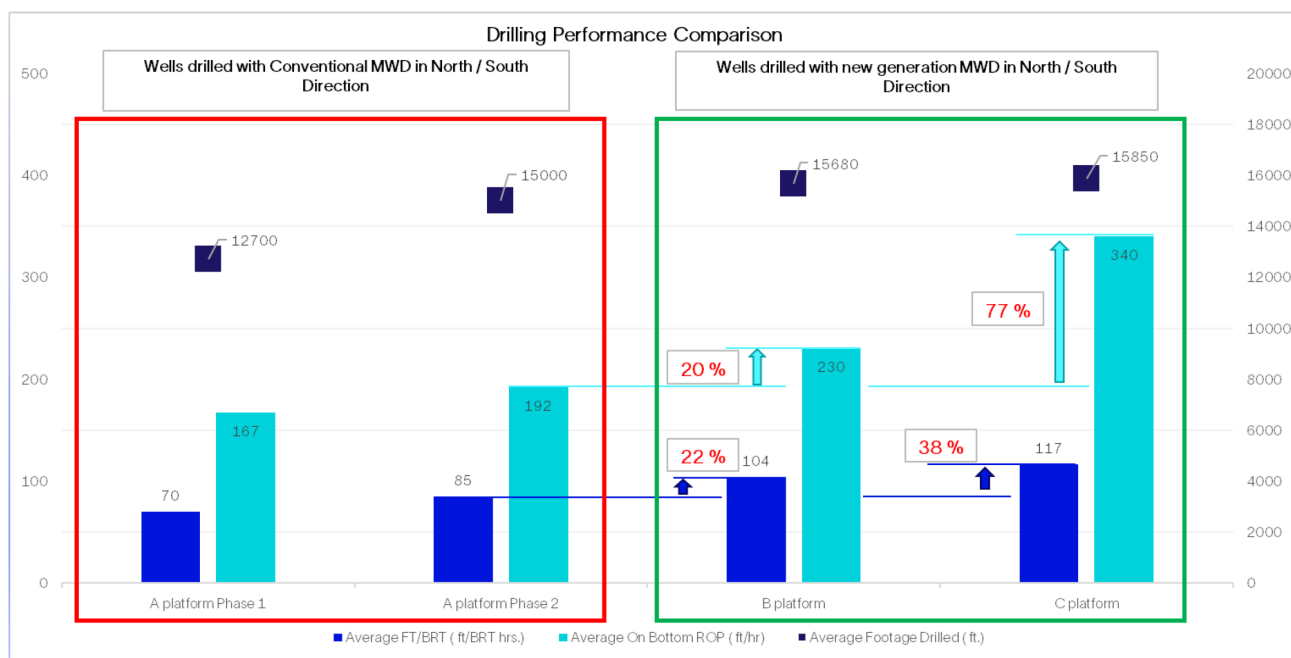


Figure 5—Drilling Performance Comparison between platforms that were drilled with conventional MWD tool (on the left) and the new generation MWD tool with DDS (on the right).

Building on these results, lessons learned were applied to subsequent operations on other platforms, leading to even greater performance improvements:

- 77% increase in On-Bottom ROP
- 38% increase in FT/BRT hours

These gains reflect reduced need for corrective actions and more efficient drilling execution.

Wellbore Tortuosity Reduction

New enhanced telemetry, combined with real-time tortuosity monitoring, contributed to a significant reduction in wellbore tortuosity. By introducing this complex answer product direction drilling specialist has now ability to perform fast decision for proactive approach in trajectory control. As demonstrated in Figure 6, comparative analysis between wells drilled with the new generation MWD tool and previously drilled wells showed:

- 23% reduction in average tortuosity at 15,000 ft. MD
- 19% reduction at 20,000 ft. MD.

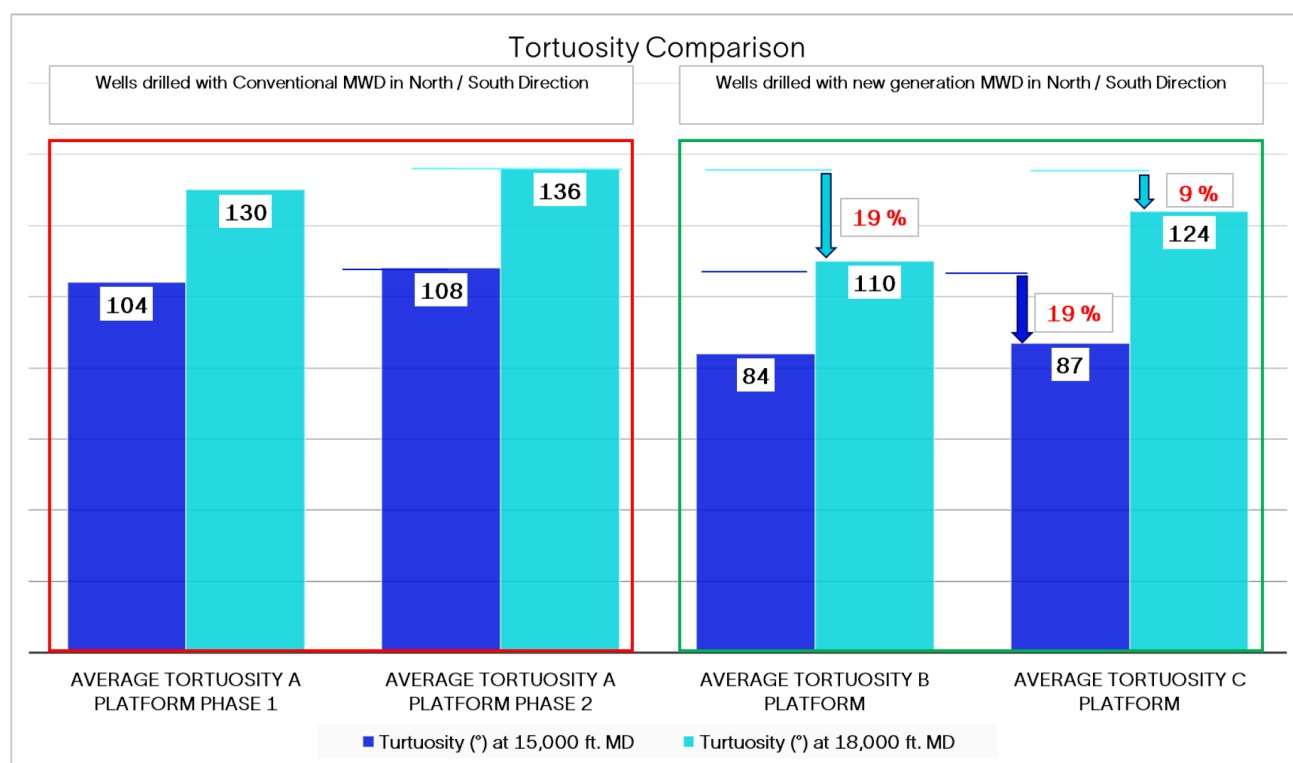


Figure 6—Tortuosity Comparison between platforms that were drilled with conventional (on the left) MWD tool and the new generation (on the right) MWD tool with DDS

This improvement directly impacted liner run success rates and allowed drilling of more extended trajectories due to the reduction in torque and drag (T&D)

These results demonstrate that the strategic deployment of new generation MWD tool with enhanced D&I sensor package and telemetry setup not only mitigates the challenges of North-South drilling but also transforms operational workflows—paving the way for smarter, more efficient, and safer drilling practices.

Conclusion

The application of dynamic 6-axis surveys mitigates many of the issues associated with the static Azimuth accuracy, particularly during the horizontal section drilling in North-South direction. The availability of

high-quality survey data in real-time while BHA is rotating, greatly improves control over well path optimization and reduces both drilling risks and time. Field data demonstrates that DDS is better than the Conventional Continuous method because of its ability to avoid large Azimuth errors due to inaccuracies of cross-axis magnetic readings.

Implementing DDS, however, requires a detailed oriented approach to ensure its effectiveness across diverse operational conditions. First, thorough advance planning is essential, particularly in critical zones where the North-South orientation can amplify azimuth errors. This involves not only identifying high-risk areas but also defining specific operational parameters that account for the limitations of conventional surveying techniques. Secondly, the real-time data transmission process must be reviewed and enhanced to accommodate a more comprehensive, holistic approach to data acquisition. This includes upgrading existing systems to handle the continuous inflow of high-resolution data and ensuring seamless integration into the acquisition pipeline (downhole data, surface data and analytical processing). Additionally, data validation and quality control protocols need to be strengthened to minimize the risks of erroneous inputs that could compromise wellbore positioning. Thirdly, the drilling crew must adapt operational workflows to incorporate high-frequency data. This data will allow early identification of deviation trends, enabling proactive adjustments to the well path before significant errors accumulate.

These results provide important guidance for future wells. Full-scale adoption of DDS is expected to reduce tortuosity, enhance geosteering accuracy, and increase overall drilling efficiency. The next step is to integrate DDS data directly into the real-time drilling automation system, enabling instantaneous well-path adjustments. With proper implementation, these advances should deliver faster, more cost-effective, and more efficient drilling in future wells.

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